



COAL Semester Project Proposal

Modular Command-Based Operating System Shell

Group Name: Opcode Engineers

Group Members:

Muhammad Bilal- Group Leader	(241815)
Muhammad Ayan Hamdani	(241872)
Aleena Mushtaq	(241902)

Submitted To: **Mam Romana Maroof**

2nd May 2026

Nano OS – A Minimal Terminal-Based Operating System in Assembly Language

1) Introduction

Nano OS is a lightweight, terminal-based operating system developed using Assembly language. The system is designed to boot directly from a raw binary without relying on existing operating systems such as Windows or Linux. The purpose of this project is to explore low-level system design and understand how software interacts directly with hardware during the boot process.

2) Objectives

The key objectives of this project are:

- To understand the booting mechanism of a computer system
- To develop a functional bootloader
- To implement a minimal kernel for handling input and output
- To design a command-line interface (CLI)
- To integrate simple built-in programs such as a calculator and a number guessing game
- To strengthen knowledge of Assembly language and low-level programming

3) Scope of the Project

Nano OS will operate entirely in a text-based (terminal) environment without graphical user interface support. The system will allow users to enter commands and execute predefined operations. The OS will be executed in a virtual environment using an emulator such as QEMU to ensure safe testing and development.

The system will also include a modular design where multiple built-in applications (modules) are integrated within the command-based environment. These modules will be accessible through specific commands and will perform different functionalities. For example, a calculator module for basic arithmetic operations and a number guessing game for user interaction will be included. Additional modules may also be implemented to enhance the functionality and user experience of the system like Sudoku Solver, Typing speed Tester, Assembly data visualizer, Music Composer

4) Methodology

The project will follow a modular and phased development approach:

Phase 1: Bootloader Development

- Create a bootable binary (512 bytes)
- Display an initial message on the screen
- Ensure correct boot signature

Phase 2: Kernel Development

- Implement basic screen output using BIOS interrupts
- Handle keyboard input from the user
- Maintain control flow of the system

Phase 3: Command-Line Interface (CLI)

- Develop a prompt system (e.g., Nano OS)
- Capture and store user input
- Implement command parsing and execution

Phase 4: Program Integration

- Develop a simple calculator (addition, subtraction, multiplication)
- Implement a number guessing game
- Add utility commands such as help and clear

5) Tools and Technologies

- Assembly Language (x86 Architecture)
- NASM (Netwide Assembler)
- QEMU (for execution and testing)
- Text Editor / IDE (e.g., Visual Studio Code)

6) Expected Outcomes

- A bootable mini operating system (Nano OS)
- A working command-line interface
- Functional built-in programs (calculator and game)
- Improved understanding of low-level system architecture and hardware interaction

7) Significance of Project

This project provides practical exposure to system-level programming and demonstrates how operating systems function internally. It is particularly beneficial for students interested in operating systems, cybersecurity, embedded systems, and low-level software development.

8) Conclusion

Nano OS aims to demonstrate the fundamental concepts of operating systems in a simplified and educational manner. By building the system from scratch using Assembly language, this project highlights the core principles of booting, memory handling, and direct hardware interaction, forming a strong foundation for advanced system development.